

Electronic Companion for the Paper:
*Event-Triggered Holomorphic Embedding Safe Reinforcement
Learning for OPF in Integrated Electric-Gas Systems*

Documentation for Integrated Electric-Gas System Topology Benchmarks

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1 Data Sources

This document describes the data sources of the three integrated energy system subsystems used in the current experiments. The electric-network topologies are taken from MATPOWER and ACTIVSg power-system data¹; the natural-gas network topologies are taken from the GasLib benchmark data².

2 Data Structure

2.1 Column Definitions for Gas-Network Data Tables

¹Zimmerman R D, Murillo-Sánchez C E, Thomas R J. MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education[J]. IEEE Transactions on Power Systems, 2010, 26(1): 12–19.

²Schmidt M, Aßmann D, Burlacu R, et al. GasLib–A library of gas network instances[J]. Data, 2017, 2(4): 40.

Table 1: Column definitions of the gas-network data tables.

Data table	Columns	Meaning and units
Well	0–6	gas-supply node, maximum/minimum gas production (MSm ³ /h), cost coefficient, gas-source index, reference-well flag, and gas production
GenGas	0–3	unit type, gas-consumption efficiency, gas node, and electric-side bus; type 1 denotes a gas turbine and type 0 denotes a conventional generation plant
Demand	0–23	24-period gas-load multipliers; the initial values are all ones and are replaced by the prescribed operating profile
Load	0–5	gas node, maximum pressure, minimum pressure, base gas load, node-type flag, and reference-node pressure
Pipeline	0–5	pipeline inlet node, pipeline outlet node, length (m), diameter (m), K_m -related parameter, and K_q -related parameter
Comp	0–14	inlet node, outlet node, type, compression factor, α , k_1 , k_2 , compression-ratio limits, electric-power limits (only for electric-driven compressors), electric-side bus, loss, efficiency, and current ratio
P2G	0–6	electric-side load bus, gas node, efficiency, current power (MW), base capacity (MVA), upper power limit (MW), and lower power limit (MW)

3 Benchmark Overview

Table 2: Sizes of the three IEGS benchmarks.³

Case	Power buses count	Power loads count	Gen. units	GT units	Gas nodes count	Gas loads count	Pipes branches	Comp units	P2G units
P14_G11	14	11	5	2	11	3	8	2	1
P57_G40	57	42	7	2	40	29	39	6	1
P200_G135	200	108	49	6	135	99	141	29	9

4 Calculation of Pipeline Coefficients K_m and K_q

For ordinary pipelines, the dynamic line-pack term and the steady-state flow term are characterized by coefficients K_m and K_q , respectively. For ordinary pipeline ℓ , let X_ℓ denote the pipeline length (m), D_ℓ the pipe diameter (m), $Z = 0.8$ the compressibility factor, $T = 281.15$ K the operating temperature, $G_s = 0.6106$ the relative density of natural gas, and $\epsilon = 0.05$ the pipe roughness parameter. Gas pressure is expressed in bar and gas flow is expressed in MSm³/h.

To clarify the origin of these two coefficients, consider the simplified transient model of a horizontal gas transmission pipeline. After neglecting elevation and inertial terms, the continuity and momentum equations can be written as

$$\frac{\partial(\rho v)}{\partial x} + \frac{\partial \rho}{\partial t} = 0, \quad (1a)$$

$$\frac{\partial p}{\partial x} + \frac{f_\ell^{\text{fr}} \rho v |v|}{2D_\ell} = 0, \quad (1b)$$

³GT denotes gas turbine; GP denotes generation plant; P2G denotes power-to-gas; Comp denotes compressor.

where ρ is the gas density, v is the in-pipe gas velocity, p is pressure, and f_ℓ^{fr} is the friction factor. Combining the gas equation of state with the standard-state volumetric-flow definition gives

$$p = \rho RTZ, \quad (2a)$$

$$\tilde{f}_\ell = \frac{\pi D_\ell^2}{4\rho_0} \rho v, \quad (2b)$$

where R is the gas constant, ρ_0 is the standard-state gas density, and $\tilde{f}_\ell$ is the standard-state volumetric flow. The pressure-flow form of the pipeline equations is then obtained as

$$\frac{\partial \tilde{f}_\ell}{\partial x} + \frac{\pi D_\ell^2}{4\rho_0 RTZ} \frac{\partial p}{\partial t} = 0, \quad (3a)$$

$$p \frac{\partial p}{\partial x} + \left(\frac{4}{\pi}\right)^2 \frac{f_\ell^{\text{fr}} RTZ \rho_0^2}{2D_\ell^5} \tilde{f}_\ell |\tilde{f}_\ell| = 0. \quad (3b)$$

The factor $\pi/4$ originates from the circular cross-sectional area $A_\ell = \pi D_\ell^2/4$. In the flow equation, because velocity is expressed as $v \propto \tilde{f}_\ell/A_\ell$ and the friction term contains $v|v|$, the squared-area coefficient associated with $(\pi/4)^2$ appears.

Discretizing pipeline $\ell = (i, j)$ over its length X_ℓ , and representing the pipe state by the endpoint-pressure average $\bar{p}_{\ell,t} = (p_{i,t} + p_{j,t})/2$ and the average flow $\tilde{f}_{\ell,t} = (F_{\ell,t}^{\text{in}} + F_{\ell,t}^{\text{out}})/2$, yields the line-pack term and the Weymouth-type flow term:

$$m_{\ell,t} = \Phi_\ell^m \frac{\pi X_\ell D_\ell^2}{4RTZ\rho_0} \bar{p}_{\ell,t}, \quad (4a)$$

$$m_{\ell,t} = m_{\ell,t-1} + F_{\ell,t}^{\text{in}} - F_{\ell,t}^{\text{out}}, \quad (4b)$$

$$\tilde{f}_{\ell,t} |\tilde{f}_{\ell,t}| = \Phi_\ell^q \left(\frac{\pi}{4}\right)^2 \frac{D_\ell^5}{X_\ell f_\ell^{\text{fr}} RTZ \rho_0^2} (p_{i,t}^2 - p_{j,t}^2). \quad (4c)$$

Here, Φ_ℓ^m and Φ_ℓ^q collect the conversions among pressure, time, and standard-volume units. If $\Sigma_{\ell,t} = F_{\ell,t}^{\text{in}} + F_{\ell,t}^{\text{out}}$ is introduced, then $\tilde{f}_{\ell,t} = \Sigma_{\ell,t}/2$. Therefore, the Weymouth relation in average-flow form can equivalently be written as $\Sigma_{\ell,t} |\Sigma_{\ell,t}| = 4K_{q,\ell} (p_{i,t}^2 - p_{j,t}^2)$.

Under the engineering units adopted in this document, the above line-pack and flow terms are folded into two tabulated coefficients, $K_{m,\ell}$ and $K_{q,\ell}$. The friction correction term is first computed as⁴

$$\Lambda_\ell = \left[2 \log_{10} \left(\frac{3.7 D_\ell}{\epsilon} \right) \right]^2, \quad (5)$$

where Λ_ℓ can be interpreted as an approximate expression for the reciprocal of the Darcy friction factor. After combining $1/f_\ell^{\text{fr}}$, standard-state parameters, and unit conversions, the pipeline coefficients are written as

$$K_{m,\ell} = C_m \frac{X_\ell D_\ell^2}{TZ}, \quad K_{q,\ell} = C_q \frac{D_\ell^5 \Lambda_\ell}{ZTX_\ell G_s}. \quad (6)$$

The folded constants used in this document are

$$C_m = 2.119 \times 10^{-7}, \quad C_q = 96.074830 \times 10^{-15}. \quad (7)$$

These two numerical constants result from combining standard-state parameters, pressure units, time units, and volume scaling. Taking the standard temperature as $T_0 = 273.15$ K, the standard pressure as $p_0 = 1.01325$ bar, and the air density as $\rho_{\text{air}} = 1.2922$ kg/m³, the line-pack constant can be obtained from the ideal-gas equation of state and the circular pipe cross section:

$$C_m \simeq \frac{\pi T_0}{4 p_0} \times 10^{-9} = \frac{\pi}{4} \frac{273.15}{1.01325} \times 10^{-9} \approx 2.12 \times 10^{-7}, \quad (8)$$

⁴Humpola J. Gas network optimization by MINLP[M]. Logos Verlag Berlin GmbH, 2017; De Wolf D, Smeers Y. The gas transmission problem solved by an extension of the simplex algorithm[J]. Management Science, 2000, 46(11): 1454–1465.

where 10^{-9} accounts for the consistent conversion from standard volume to million standard cubic meters and for the length and area scales.

The flow-term constant C_q follows from the Weymouth-type pipeline-flow relation. When the squared-flow term is expressed in MSm^3/h and the squared-pressure term in bar^2 , the part independent of pipe diameter, length, and friction can be written as

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{T_0}{p_0 \rho_{\text{air}}} s_q, \quad (9)$$

where $(\pi/4)^2$ originates from the square of the circular cross-sectional area, 10^5 converts the pressure scale between bar and Pa, $(3600)^2$ converts squared flow from seconds to hours, $T_0/(p_0 \rho_{\text{air}})$ converts between standard-state volume and mass density, and s_q collects the MSm^3 volume scale, the squared-pressure scale, and the length-diameter unit convention adopted by the gas-network data. Under the unit system used here,

$$s_q = 5.7606303 \times 10^{-28}, \quad (10)$$

therefore

$$C_q = \left(\frac{\pi}{4}\right)^2 10^5 (3600)^2 \frac{273.15}{1.01325 \times 1.2922} \times 5.7606303 \times 10^{-28} \approx 96.074830 \times 10^{-15}. \quad (11)$$

Thus, the tabulated K_m and K_q values are the pipeline coefficients obtained from the line-pack and Weymouth flow terms under the above unit system, given X_ℓ , D_ℓ , and Λ_ℓ .

It should be distinguished that the energy-volume conversion coefficients for electric-gas coupling elements are based on a separate set of EIA energy-unit data⁵, which is used to relate per-unit electric power to natural-gas volume. The simplified conversion adopted here is

$$1 \text{ MWh} \approx 3412.1416 \text{ ft}^3 \times 0.02831685 \approx 96.6 \text{ m}^3, \quad (12)$$

Therefore, when 100 MWh is used as the electric-power base,

$$\alpha_0 = \frac{96.6 \times 100}{10^6} = 9.66 \times 10^{-3} \text{ MSm}^3/(100 \text{ MWh}). \quad (13)$$

The electric-gas coupling coefficients of gas turbines, P2G units, and compressors are then obtained from α_0 by multiplying or dividing by the corresponding efficiency parameters.

5 P14_G11

5.1 Construction Summary

P14_G11 consists of the IEEE 14-bus power network and an 11-node gas network. The power network is constructed using `pandapower.networks.case14()`; the 11-node gas network is a purpose-built small synthetic pipeline network for testing, containing three gas wells, three load nodes, eight ordinary pipelines, two compressors, and one P2G unit.

5.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

⁵EIA – Units & Calculators Explained, <https://www.eia.gov/energyexplained/units-and-calculators/>.

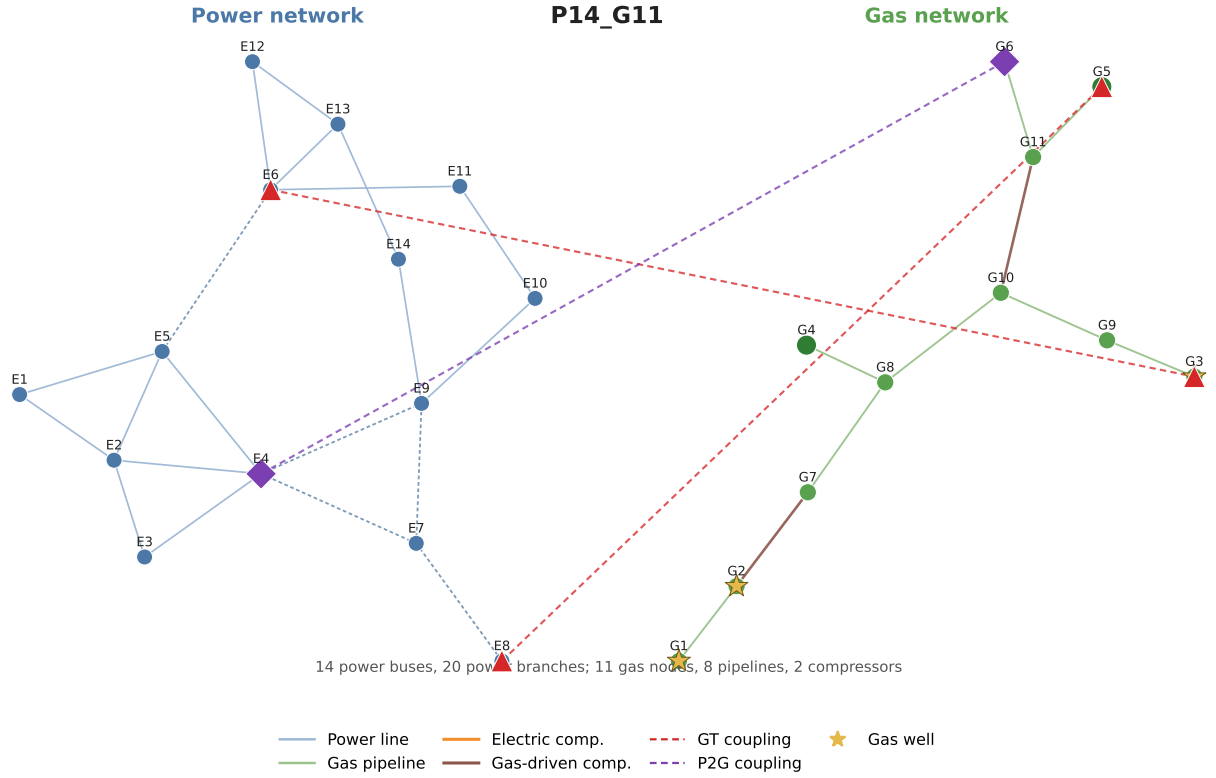


Figure 1: P14_G11 topology of the integrated electric-gas system. In the figure, E denotes an electric bus and G denotes a gas-network node.

Table 3: P14_G11 Gas-well data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	0.16	0	228.566	1	0
2	0.115	0	190.472	0	0
3	1	0	355.548	0	0

Gas["Load"] : Gas-Network Node Data

Table 4: P14_G11 Gas-network node data.

No.	Node	$p_{\max}$ bar	$p_{\min}$ bar	Base load MSm ³ /h	Type	Pressure bar
1	1	70	40	0	1	50
2	2	70	40	0	3	0
3	3	70	40	0	3	0
4	4	70	40	0.1	2	0
5	5	60	40	0.12	2	0
6	6	60	40	0.08	2	0
7	7	70	40	0	0	0

No.	Node	$p_{\max}$ bar	$p_{\min}$ bar	Base load MSm^3 /h	Type	Pressure bar
8	8	70	40	0	0	0
9	9	70	40	0	0	0
10	10	70	40	0	0	0
11	11	70	40	0	0	0

Gas["Pipeline"]: Gas-network Pipeline Data

Table 5: P14_G11 Gas-network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	Diameter m	K_m MSm^3 h^{-1} Pa^{-1}	K_q MSm^6 h^{-2} bar^{-2}
1	1	2	55000	0.5	0.0152	0.000209845
2	7	8	55000	0.5	0.0188	4.74367e-05
3	3	9	55000	0.5	0.0129	6.89989e-05
4	8	4	55000	0.5	0.0118	7.58988e-05
5	8	10	55000	0.5	0.0271	0.000118038
6	9	10	55000	0.5	0.0129	6.89989e-05
7	11	5	55000	0.5	0.0188	4.74367e-05
8	11	6	55000	0.5	0.0188	4.74367e-05

Table 6: P14_G11 Generators and GT Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r	V p.u.
1	1	GP	–	–	232.40	0.00	332.40	0.00	12.00	1.060
2	2	GP	–	–	40.00	14.00	140.00	-40.00	50.00	1.045
3	3	GP	–	–	0.00	14.00	100.00	0.00	40.00	1.010
4	6	GT	3	0.55	0.00	0.00	100.00	-6.00	24.00	1.070
5	8	GT	5	0.45	0.00	0.00	100.00	-6.00	24.00	1.090

Table 7: P14_G11 P2G Data.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	4	6	0.45	0.00	5.00

Table 8: P14_G11 Compressor Data.

No.	Gas inlet	Gas outlet	Type	Power bus	Ratio p.u.	Elec. power range MW
1	2	7	Gas	6	1.00–1.20	–
2	10	11	Gas	6	1.00–1.20	–

6 P57_G40

6.1 Construction Summary

P57_G40 consists of the IEEE 57-bus power network and a 40-node gas network. This 40-node gas network roughly represents a segment of Germany's low-calorific gas transmission network serving the Rhine-Main-Ruhr region. The network contains three gas wells, 29 load nodes, 39 ordinary pipelines, six compressors, and one P2G unit.

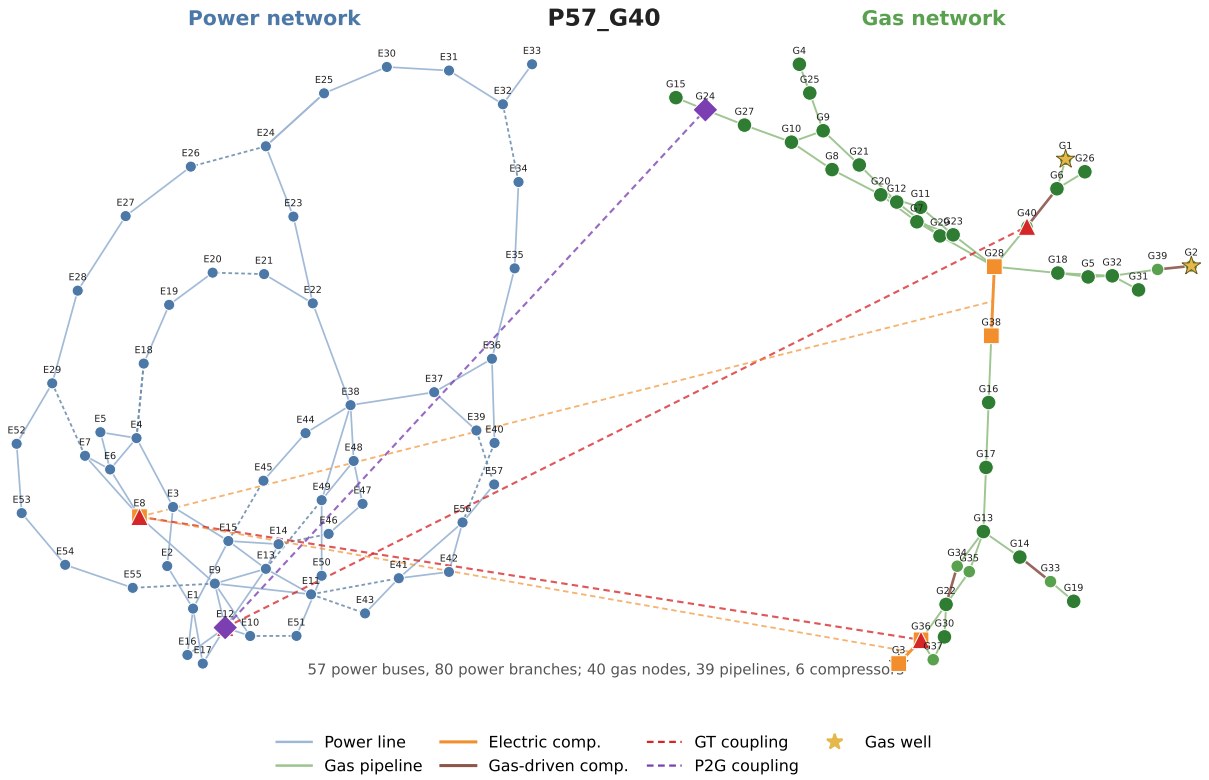


Figure 2: P57_G40 topology of the integrated electric-gas system. In the figure, E denotes an electric bus and G denotes a gas-network node.

6.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

Gas["Well"]: Gas-Well Data

Table 9: P57_G40 Gas-well data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	2.5	-1	2692.34	1	0
2	1.5	0	2243.61	0	0
3	1.5	0	2243.61	0	0

Gas["Load"]: Gas-Network Node Data

Table 10: P57_G40 Gas-network node data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
1	81.0132	68	0	1	74	21	81.0132	30	0.075	2	0
2	81.0132	30	0	3	0	22	81.0132	30	0.075	2	0
3	81.0132	30	0	3	0	23	81.0132	30	0.075	2	0
4	81.0132	30	0.075	2	0	24	81.0132	30	0.075	2	0
5	81.0132	30	0.075	2	0	25	81.0132	30	0.075	2	0
6	81.0132	30	0.075	2	0	26	81.0132	30	0.075	2	0
7	81.0132	30	0.075	2	0	27	81.0132	30	0.075	2	0
8	81.0132	30	0.075	2	0	28	81.0132	30	0.075	2	0
9	81.0132	30	0.075	2	0	29	81.0132	30	0.075	2	0
10	81.0132	30	0.075	2	0	30	81.0132	30	0.075	2	0
11	81.0132	30	0.075	2	0	31	81.0132	30	0.075	2	0
12	81.0132	30	0.075	2	0	32	81.0132	30	0.075	2	0
13	81.0132	30	0.075	2	0	33	81.0132	30	0	0	0
14	81.0132	30	0.075	2	0	34	81.0132	30	0	0	0
15	81.0132	30	0.075	2	0	35	81.0132	30	0	0	0
16	81.0132	30	0.075	2	0	36	81.0132	30	0	0	0
17	81.0132	30	0.075	2	0	37	81.0132	30	0	0	0
18	81.0132	30	0.075	2	0	38	81.0132	30	0	0	0
19	81.0132	30	0.075	2	0	39	81.0132	30	0	0	0
20	81.0132	30	0.075	2	0	40	81.0132	30	0	0	0

Gas["Pipeline"]: Gas-network Pipeline Data

Table 11: P57_G40 Gas-network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²
1	1	6	13071	1	0.0152	0.000209845	21	20	7	10586	0.6	0.0188	4.74367e-05
2	33	19	38447	0.8	0.0188	4.74367e-05	22	20	11	10452	0.6	0.0129	6.89989e-05
3	38	16	21558	1	0.0129	6.89989e-05	23	6	26	12397	0.8	0.0118	7.58988e-05
4	16	17	6998	1	0.0118	7.58988e-05	24	11	23	19303	0.6	0.0271	0.000118038
5	17	13	58219	0.8	0.0271	0.000118038	25	28	23	66037	0.6	0.0129	6.89989e-05
6	28	29	43345	0.8	0.0129	6.89989e-05	26	28	18	18969	1	0.0188	4.74367e-05
7	29	12	16579	0.6	0.0188	4.74367e-05	27	18	32	36061	0.8	0.0188	4.74367e-05
8	12	21	10023	0.6	0.0188	4.74367e-05	28	32	31	22224	0.8	0.0271	0.000118038
9	29	7	35219	0.6	0.0271	0.000118038	29	32	5	31180	0.8	0.0188	4.74367e-05
10	7	23	20322	0.6	0.0188	4.74367e-05	30	5	18	12767	1	0.0188	4.74367e-05
11	21	9	32868	0.8	0.0188	4.74367e-05	31	32	39	32921	0.8	0.0188	4.74367e-05
12	28	40	47488	0.8	0.0188	4.74367e-05	32	36	22	49866	0.8	0.0188	4.74367e-05
13	9	10	3803	0.6	0.0188	4.74367e-05	33	22	35	3479	0.8	0.0188	4.74367e-05
14	9	25	39036	1.8	0.0188	4.74367e-05	34	36	37	3418	1	0.0188	4.74367e-05
15	10	27	38660	1.8	0.0188	4.74367e-05	35	30	37	32449	1	0.0188	4.74367e-05
16	25	4	18018	1.8	0.0188	4.74367e-05	36	30	22	26427	0.8	0.0188	4.74367e-05
17	27	24	3068	1.8	0.0188	4.74367e-05	37	13	14	18137	1	0.0188	4.74367e-05
18	24	15	12016	1.8	0.0188	4.74367e-05	38	13	34	32528.5	0.8	0.0188	4.74367e-05
19	10	8	14043	0.4	0.0188	4.74367e-05	39	13	35	32766	0.8	0.0188	4.74367e-05
20	8	20	20635	0.6	0.0152	0.000209845							

Table 12: P57_G40 Generators and GT Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r	V p.u.
1	1	GP	–	–	128.90	0.00	575.88	-140.00	200.00	1.040
2	2	GP	–	–	0.00	0.00	100.00	-17.00	50.00	1.010
3	3	GP	–	–	40.00	0.00	140.00	-10.00	60.00	0.985
4	6	GP	–	–	0.00	0.00	100.00	-8.00	25.00	0.980
5	8	GT	36	0.55	450.00	0.00	550.00	-140.00	200.00	1.005
6	9	GP	–	–	0.00	0.00	100.00	-3.00	9.00	0.980
7	12	GT	40	0.45	310.00	0.00	410.00	-150.00	155.00	1.015

Table 13: P57_G40 P2G Data.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	12	24	0.45	70.00	150.00

Table 14: P57_G40 Compressor Data.

No.	Gas inlet	Gas outlet	Type	Power bus	Ratio p.u.	Elec. power range MW
1	38	28	Electric	8	1.00–1.20	0–800
2	14	33	Gas	8	1.00–1.20	–
3	22	34	Gas	12	1.00–1.20	–
4	3	36	Electric	8	1.00–1.20	0–800
5	2	39	Gas	8	1.00–1.20	–
6	6	40	Gas	12	1.00–1.20	–

7 P200_G135

7.1 Construction Summary

P200_G135 consists of the ACTIVSg200 power network and a 135-node gas network. This 135-node gas network roughly represents a portion of Germany's high-calorific gas transmission network. The network contains six GT units, nine P2G units, and 29 compressors.

7.2 Gas-Side Data

This subsection reports the gas-side benchmark tables `Gas["Well"]`, `Gas["Load"]`, and `Gas["Pipeline"]`. The table `Gas["Load"]` corresponds to the gas-network node data. In the node-type flag, 1 denotes a reference node, 2 a gas-load node, 3 a gas-source node, and 0 an ordinary junction node.

`Gas["Well"]`: Gas-Well Data

Table 15: P200_G135 Gas-well data.

No.	$G_{\max}$ MSm ³ /h	$G_{\min}$ MSm ³ /h	Cost coeff.	Ref. well	Output MSm ³ /h
1	15	-15	4140	1	0
2	2	0	3450	0	0
3	2	0	3450	0	0
4	2	0	3450	0	0
5	2	0	3450	0	0
6	2	0	3450	0	0

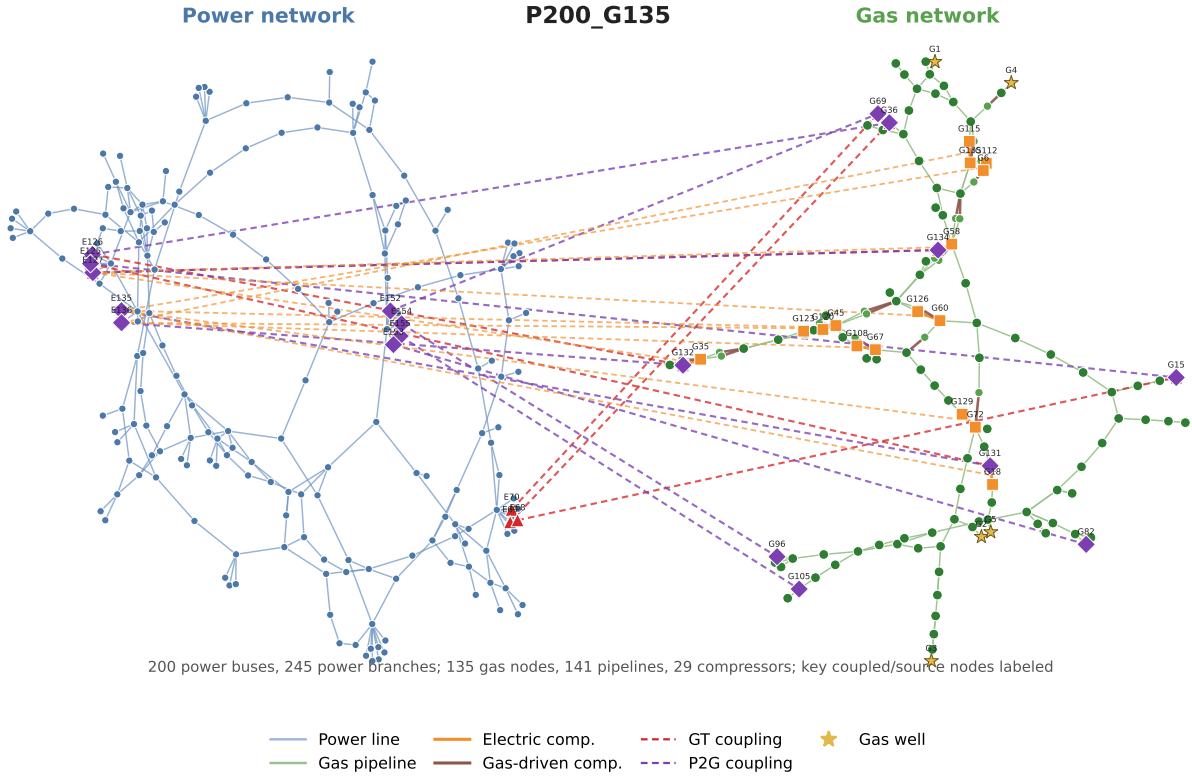


Figure 3: P200_G135 topology of the integrated electric-gas system. In the figure, E denotes an electric bus and G denotes a gas-network node; only coupled nodes and gas-source nodes are labeled to avoid visual clutter.

Gas["Load"] : Gas-Network Node Data

Table 16: P200_G135 Gas-network node data.

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
1	75	60	0	1	60	69	81.0132	30	0.04	2	0
2	81.0132	30	0	3	0	70	81.0132	30	0.04	2	0
3	81.0132	30	0	3	0	71	81.0132	30	0.04	2	0
4	81.0132	30	0	3	0	72	81.0132	30	0.04	2	0
5	81.0132	30	0	3	0	73	81.0132	30	0.04	2	0
6	81.0132	30	0	3	0	74	81.0132	30	0.04	2	0
7	81.0132	30	0.04	2	0	75	81.0132	30	0.04	2	0
8	81.0132	30	0.04	2	0	76	81.0132	30	0.04	2	0
9	81.0132	30	0.04	2	0	77	81.0132	30	0.04	2	0
10	81.0132	30	0.04	2	0	78	81.0132	30	0.04	2	0
11	81.0132	30	0.04	2	0	79	81.0132	30	0.04	2	0
12	81.0132	30	0.04	2	0	80	81.0132	30	0.04	2	0
13	81.0132	30	0.04	2	0	81	81.0132	30	0.04	2	0
14	81.0132	30	0.04	2	0	82	81.0132	30	0.04	2	0
15	81.0132	30	0.04	2	0	83	81.0132	30	0.04	2	0

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
16	81.0132	30	0.04	2	0	84	81.0132	30	0.04	2	0
17	81.0132	30	0.04	2	0	85	81.0132	30	0.04	2	0
18	81.0132	30	0.04	2	0	86	81.0132	30	0.04	2	0
19	81.0132	30	0.04	2	0	87	81.0132	30	0.04	2	0
20	81.0132	30	0.04	2	0	88	81.0132	30	0.04	2	0
21	81.0132	30	0.04	2	0	89	81.0132	30	0.04	2	0
22	81.0132	30	0.04	2	0	90	81.0132	30	0.04	2	0
23	81.0132	30	0.04	2	0	91	81.0132	30	0.04	2	0
24	81.0132	30	0.04	2	0	92	81.0132	30	0.04	2	0
25	81.0132	30	0.04	2	0	93	81.0132	30	0.04	2	0
26	81.0132	30	0.04	2	0	94	81.0132	30	0.04	2	0
27	81.0132	30	0.04	2	0	95	81.0132	30	0.04	2	0
28	81.0132	30	0.04	2	0	96	81.0132	30	0.04	2	0
29	81.0132	30	0.04	2	0	97	81.0132	30	0.04	2	0
30	81.0132	30	0.04	2	0	98	81.0132	30	0.04	2	0
31	81.0132	30	0.04	2	0	99	81.0132	30	0.04	2	0
32	81.0132	30	0.04	2	0	100	81.0132	30	0.04	2	0
33	81.0132	30	0.04	2	0	101	81.0132	30	0.04	2	0
34	81.0132	30	0.04	2	0	102	81.0132	30	0.04	2	0
35	81.0132	30	0.04	2	0	103	81.0132	30	0.04	2	0
36	81.0132	30	0.04	2	0	104	81.0132	30	0.04	2	0
37	81.0132	30	0.04	2	0	105	81.0132	30	0.04	2	0
38	81.0132	30	0.04	2	0	106	81.0132	30	0	0	0
39	81.0132	30	0.04	2	0	107	81.0132	30	0	0	0
40	81.0132	30	0.04	2	0	108	81.0132	30	0	0	0
41	81.0132	30	0.04	2	0	109	81.0132	30	0	0	0
42	81.0132	30	0.04	2	0	110	81.0132	30	0	0	0
43	81.0132	30	0.04	2	0	111	81.0132	30	0	0	0
44	81.0132	30	0.04	2	0	112	81.0132	30	0	0	0
45	81.0132	30	0.04	2	0	113	81.0132	30	0	0	0
46	81.0132	30	0.04	2	0	114	81.0132	30	0	0	0
47	81.0132	30	0.04	2	0	115	81.0132	30	0	0	0
48	81.0132	30	0.04	2	0	116	81.0132	30	0	0	0
49	81.0132	30	0.04	2	0	117	81.0132	30	0	0	0
50	81.0132	30	0.04	2	0	118	81.0132	30	0	0	0
51	81.0132	30	0.04	2	0	119	81.0132	30	0	0	0
52	81.0132	30	0.04	2	0	120	81.0132	30	0	0	0
53	81.0132	30	0.04	2	0	121	81.0132	30	0	0	0
54	81.0132	30	0.04	2	0	122	81.0132	30	0	0	0

No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar	No.	$p_{\max}$ bar	$p_{\min}$ bar	Load MSm ³ /h	Type	p_{ref} bar
55	81.0132	30	0.04	2	0	123	81.0132	30	0	0	0
56	81.0132	30	0.04	2	0	124	81.0132	30	0	0	0
57	81.0132	30	0.04	2	0	125	81.0132	30	0	0	0
58	81.0132	30	0.04	2	0	126	81.0132	30	0	0	0
59	81.0132	30	0.04	2	0	127	81.0132	30	0	0	0
60	81.0132	30	0.04	2	0	128	81.0132	30	0	0	0
61	81.0132	30	0.04	2	0	129	81.0132	30	0	0	0
62	81.0132	30	0.04	2	0	130	81.0132	30	0	0	0
63	81.0132	30	0.04	2	0	131	81.0132	30	0	0	0
64	81.0132	30	0.04	2	0	132	81.0132	30	0	0	0
65	81.0132	30	0.04	2	0	133	81.0132	30	0	0	0
66	81.0132	30	0.04	2	0	134	81.0132	30	0	0	0
67	81.0132	30	0.04	2	0	135	81.0132	30	0	0	0
68	81.0132	30	0.04	2	0						

Gas["Pipeline"]: Gas-network Pipeline Data

Table 17: P200_G135 Gas-network Pipeline Data.

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²
1	3	28	50304.8	1.2	0.0152	0.000209845	72	39	60	56432.5	1.2	0.0188	4.74367e-05
2	131	25	50170.3	1.2	0.0188	4.74367e-05	73	108	63	79066.7	1.2	0.0188	4.74367e-05
3	90	91	17722	1.2	0.0129	6.89989e-05	74	107	45	84957.9	1.2	0.0188	4.74367e-05
4	90	91	17722	1.2	0.0118	7.58988e-05	75	63	45	99697	1.2	0.0188	4.74367e-05
5	91	38	27576	1.2	0.0271	0.000118038	76	2	26	42281.5	1.2	0.0188	4.74367e-05
6	90	7	50039.1	1.2	0.0129	6.89989e-05	77	45	56	92797.1	1.2	0.0188	4.74367e-05
7	5	2	35005.3	1.2	0.0188	4.74367e-05	78	120	42	24981.3	1.2	0.0188	4.74367e-05
8	28	51	35640.1	1.2	0.0188	4.74367e-05	79	121	42	25743.4	1.2	0.0188	4.74367e-05
9	51	38	12536.2	1.2	0.0271	0.000118038	80	45	124	84458.2	1.2	0.0188	4.74367e-05
10	101	105	60939.7	1.2	0.0188	4.74367e-05	81	45	125	84946.7	1.2	0.0188	4.74367e-05
11	105	98	64597.2	1.2	0.0188	4.74367e-05	82	126	31	80621.8	1.2	0.0188	4.74367e-05
12	98	99	23354.6	1.2	0.0188	4.74367e-05	83	127	31	80904.3	1.2	0.0188	4.74367e-05
13	71	7	61293	1.2	0.0188	4.74367e-05	84	122	46	83083	1.2	0.0152	0.000209845
14	99	30	43259.3	1.2	0.0188	4.74367e-05	85	123	46	83957.7	1.2	0.0188	4.74367e-05
15	30	86	39290.8	1.2	0.0188	4.74367e-05	86	46	64	13264.2	1.2	0.0129	6.89989e-05
16	86	87	38503.4	1.2	0.0188	4.74367e-05	87	26	78	49223.1	1.2	0.0118	7.58988e-05
17	87	80	65028.4	1.2	0.0188	4.74367e-05	88	46	64	13264.2	1.2	0.0271	0.000118038

No.	Pipeline inlet	Pipeline outlet	Pipeline Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²	No.	Pipeline inlet	Pipeline outlet	Pipeline Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²
18	96	17	81181.6	1.2	0.0188	4.74367e-05	89	110	35	87864.1	1.2	0.0129	6.89989e-05
19	82	81	6510.82	1.2	0.0188	4.74367e-05	90	109	35	87454.8	1.2	0.0188	4.74367e-05
20	81	89	23219.5	1.2	0.0152	0.000209845	91	132	70	33438.2	1.2	0.0188	4.74367e-05
21	89	94	69525.7	1.2	0.0188	4.74367e-05	92	133	70	33473.9	1.2	0.0271	0.000118038
22	89	94	69525.7	1.2	0.0129	6.89989e-05	93	39	53	42519.4	1.2	0.0188	4.74367e-05
23	94	93	29067.2	1.2	0.0118	7.58988e-05	94	53	68	21281.5	1.2	0.0188	4.74367e-05
24	71	73	35220.9	1.2	0.0271	0.000118038	95	39	59	37246.5	1.2	0.0188	4.74367e-05
25	93	80	73550.2	1.2	0.0129	6.89989e-05	96	59	58	63409.2	1.2	0.0188	4.74367e-05
26	94	104	46081.6	1.2	0.0188	4.74367e-05	97	31	32	13796.9	1.2	0.0188	4.74367e-05
27	104	88	49928.8	1.2	0.0188	4.74367e-05	98	5	78	10397.6	1.2	0.0188	4.74367e-05
28	94	102	85858	1.2	0.0271	0.000118038	99	31	32	13796.9	1.2	0.0188	4.74367e-05
29	94	102	85858	1.2	0.0188	4.74367e-05	100	31	32	13796.9	1.2	0.0188	4.74367e-05
30	102	11	10634.5	1.2	0.0188	4.74367e-05	101	31	40	14097.1	1.2	0.0188	4.74367e-05
31	102	11	10634.5	1.2	0.0188	4.74367e-05	102	31	40	14097.1	1.2	0.0188	4.74367e-05
32	80	66	68792.3	1.2	0.0188	4.74367e-05	103	32	116	81461.5	1.2	0.0188	4.74367e-05
33	102	85	37563.1	1.2	0.0188	4.74367e-05	104	32	117	82408.8	1.2	0.0129	6.89989e-05
34	85	83	74204.5	1.2	0.0188	4.74367e-05	105	32	34	47648.2	1.2	0.0118	7.58988e-05
35	73	21	44307.6	1.2	0.0188	4.74367e-05	106	34	134	37524.1	1.2	0.0271	0.000118038
36	83	79	34864.5	1.2	0.0188	4.74367e-05	107	58	55	55637.1	1.2	0.0129	6.89989e-05
37	97	79	10819	1.2	0.0188	4.74367e-05	108	58	118	89446.2	1.2	0.0188	4.74367e-05
38	97	103	13882.6	1.2	0.0188	4.74367e-05	109	24	78	6844.75	1.2	0.0188	4.74367e-05
39	103	14	59093.6	1.2	0.0188	4.74367e-05	110	58	119	89179.2	1.2	0.0271	0.000118038
40	79	92	74213	1.2	0.0188	4.74367e-05	111	55	43	48033.2	1.2	0.0188	4.74367e-05
41	92	9	10380	1.2	0.0188	4.74367e-05	112	55	47	19465.6	1.2	0.0188	4.74367e-05
42	9	100	39573.1	1.2	0.0188	4.74367e-05	113	47	33	93300.9	1.2	0.0188	4.74367e-05
43	100	15	25443.5	1.2	0.0188	4.74367e-05	114	47	57	31776.6	1.2	0.0188	4.74367e-05
44	92	13	81583.8	1.2	0.0188	4.74367e-05	115	57	111	75196.4	1.2	0.0188	4.74367e-05
45	13	68	49535.2	1.2	0.0152	0.000209845	116	57	135	61179.7	1.2	0.0188	4.74367e-05
46	7	65	60573.5	1.2	0.0188	4.74367e-05	117	57	135	61179.7	1.2	0.0188	4.74367e-05
47	12	27	19740	1.2	0.0129	6.89989e-05	118	135	112	27318.1	1.2	0.0188	4.74367e-05
48	10	12	25479.1	1.2	0.0118	7.58988e-05	119	135	113	26466.9	1.2	0.0188	4.74367e-05
49	65	66	94839.2	1.2	0.0271	0.000118038	120	18	24	22659.1	1.2	0.0188	4.74367e-05
50	66	30	89395.5	1.2	0.0129	6.89989e-05	121	114	62	37046	1.2	0.0188	4.74367e-05
51	30	84	67770.9	1.2	0.0188	4.74367e-05	122	115	62	37887.8	1.2	0.0188	4.74367e-05
52	84	17	10478.4	1.2	0.0188	4.74367e-05	123	62	54	51079.8	1.2	0.0188	4.74367e-05
53	17	8	9209.88	1.2	0.0271	0.000118038	124	62	106	67517.7	1.2	0.0188	4.74367e-05
54	28	51	35640.1	1.2	0.0188	4.74367e-05	125	77	4	3000.01	1.2	0.0188	4.74367e-05
55	17	37	33603.3	1.2	0.0188	4.74367e-05	126	33	74	28202.2	1.2	0.0129	6.89989e-05

No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²	No.	Pipeline inlet	Pipeline outlet	Length m	D m	K_m MSm ³ h ⁻¹ Pa ⁻¹	K_q MSm ⁶ h ⁻² bar ⁻²
56	72	21	96065	1.2	0.0188	4.74367e-05	127	74	23	56443.9	1.2	0.0118	7.58988e-05
57	72	25	61286.2	1.2	0.0188	4.74367e-05	128	23	41	78489.8	1.2	0.0271	0.000118038
58	72	22	60740.5	1.2	0.0188	4.74367e-05	129	74	36	50533.9	1.2	0.0129	6.89989e-05
59	129	75	32476.9	1.2	0.0188	4.74367e-05	130	69	36	12358.4	1.2	0.0188	4.74367e-05
60	130	20	26313.7	1.2	0.0188	4.74367e-05	131	26	71	52638.3	1.2	0.0188	4.74367e-05
61	75	76	3274.84	1.2	0.0188	4.74367e-05	132	74	44	22865.5	1.2	0.0271	0.000118038
62	76	49	54491	1.2	0.0188	4.74367e-05	133	44	27	70219.5	1.2	0.0188	4.74367e-05
63	49	50	17417.5	1.2	0.0188	4.74367e-05	134	54	29	6799.79	1.2	0.0188	4.74367e-05
64	50	67	77016.1	1.2	0.0188	4.74367e-05	135	29	27	60895.6	1.2	0.0188	4.74367e-05
65	51	38	12536.2	1.2	0.0188	4.74367e-05	136	54	95	73579.3	1.2	0.0188	4.74367e-05
66	67	16	9158.22	1.2	0.0188	4.74367e-05	137	95	61	7053.13	1.2	0.0188	4.74367e-05
67	67	16	9158.22	1.2	0.0271	0.000118038	138	27	61	44719	1.2	0.0188	4.74367e-05
68	67	52	8869.54	1.2	0.0188	4.74367e-05	139	61	19	38054.1	1.2	0.0188	4.74367e-05
69	67	52	8869.54	1.2	0.0188	4.74367e-05	140	61	1	15990.8	1.2	0.0188	4.74367e-05
70	20	39	86829.9	1.2	0.0188	4.74367e-05	141	81	48	9159.01	1.2	0.0188	4.74367e-05
71	128	60	85824.9	1.2	0.0188	4.74367e-05							

Table 18: P200_G135 Generators and GT Units.

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA	$Q_{\max}$ MVA	V p.u.
1	49	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
2	50	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
3	51	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
4	52	GP	–	–	1.36	1.36	3.90	-0.55	2.11	1.040
5	53	GP	–	–	2.72	2.72	7.80	-1.11	4.23	1.040
6	65	GP	–	–	86.50	86.50	86.50	-21.66	32.04	1.040
7	67	GP	–	–	1.41	1.41	4.04	-0.57	2.19	1.040
8	68	GT	15	0.48	8.38	8.38	27.92	-3.41	13.01	1.040
9	69	GT	36	0.46	8.38	8.38	27.92	-3.41	13.01	1.040
10	70	GT	69	0.45	8.38	8.38	27.92	-3.41	13.01	1.040
11	71	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040
12	72	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040
13	73	GP	–	–	8.38	8.38	24.01	-3.41	13.01	1.040
14	76	GP	–	–	1.20	1.20	3.44	-0.44	2.04	1.040
15	77	GP	–	–	0.72	0.72	2.06	-0.27	1.22	1.040
16	78	GP	–	–	0.00	5.40	18.00	-2.00	9.16	1.040
17	79	GP	–	–	0.00	5.40	18.00	-2.00	9.16	1.040

No.	Power bus	Type	Gas node	Eff. p.u.	P_0 MW	$P_{\min}$ MW	$P_{\max}$ MW	$Q_{\min}$ MVA _r	$Q_{\max}$ MVA _r	V p.u.
18	90	GP	–	–	0.96	0.96	2.75	-0.36	1.63	1.040
19	91	GP	–	–	1.50	1.50	4.30	-0.55	2.55	1.040
20	92	GP	–	–	0.00	1.89	6.30	-0.70	3.21	1.040
21	94	GP	–	–	5.40	5.40	15.48	-2.20	8.39	1.040
22	104	GP	–	–	67.60	67.60	67.60	-14.26	21.09	1.040
23	105	GP	–	–	154.80	154.80	154.80	-28.51	42.17	1.040
24	114	GP	–	–	1.40	1.40	1.40	-0.24	0.36	1.040
25	115	GP	–	–	133.50	133.50	133.50	-21.60	31.95	1.040
26	125	GT	131	0.55	39.02	39.02	130.05	-15.87	60.60	1.040
27	126	GT	132	0.52	39.02	39.02	130.05	-15.87	60.60	1.040
28	127	GT	134	0.50	39.02	39.02	130.05	-15.87	60.60	1.040
29	135	GP	–	–	133.92	133.92	383.90	-54.46	208.02	1.040
30	136	GP	–	–	133.92	133.92	383.90	-54.46	208.02	1.040
31	147	GP	–	–	92.40	92.40	92.40	-14.47	21.41	1.040
32	151	GP	–	–	1.62	1.62	4.64	-0.66	2.52	1.040
33	152	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
34	153	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
35	154	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
36	155	GP	–	–	23.17	23.17	66.41	-9.42	35.99	1.040
37	161	GP	–	–	0.00	41.58	138.60	-15.38	70.55	1.040
38	164	GP	–	–	0.00	3.60	12.00	-1.33	6.11	1.040
39	165	GP	–	–	0.00	7.80	26.00	-2.89	13.23	1.040
40	166	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
41	167	GP	–	–	2.82	2.82	8.08	-1.04	4.78	1.040
42	168	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
43	169	GP	–	–	0.00	2.82	9.40	-1.04	4.78	1.040
44	170	GP	–	–	2.82	2.82	8.08	-1.04	4.78	1.040
45	182	GP	–	–	5.25	5.25	15.05	-2.14	8.16	1.040
46	183	GP	–	–	7.98	7.98	22.88	-3.25	12.40	1.040
47	189	GP	–	–	384.37	170.75	569.15	-46.67	209.45	1.040
48	196	GP	–	–	0.00	20.25	67.50	-7.49	34.36	1.040
49	197	GP	–	–	0.00	20.25	67.50	-7.49	34.36	1.040

Table 19: P200_G135 P2G Data.

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
1	135	131	0.45	304.00	336.00
2	136	132	0.45	184.00	216.00
3	127	134	0.45	184.00	216.00

No.	Power bus	Gas node	Eff. p.u.	$P_{\min}$ MW	$P_{\max}$ MW
4	125	15	0.45	100.00	140.00
5	126	36	0.45	100.00	140.00
6	152	69	0.45	100.00	140.00
7	153	82	0.45	100.00	140.00
8	154	96	0.45	100.00	140.00
9	155	105	0.45	100.00	140.00

Table 20: P200_G135 Compressor Data; E and G denote electric-driven and gas-driven compressors, respectively.

No.	In.	Out.	Type	Power bus	Ratio p.u.	Elec. power MW	No.	In.	Out.	Type	Power bus	Ratio p.u.	Elec. power MW
1	18	131	E	135	1.00–1.02	0–800	16	58	134	E	127	1.00–1.02	0–800
2	77	106	G	136	1.00–1.02	–	17	57	118	G	142	1.00–1.02	–
3	67	107	G	135	1.00–1.02	–	18	57	119	G	148	1.00–1.02	–
4	67	108	E	136	1.00–1.02	0–800	19	45	120	E	135	1.00–1.02	0–800
5	64	109	G	125	1.00–1.02	–	20	45	121	G	136	1.00–1.02	–
6	64	110	G	126	1.00–1.02	–	21	45	122	G	135	1.00–1.02	–
7	35	132	E	127	1.00–1.02	0–800	22	45	123	E	136	1.00–1.02	0–800
8	35	133	G	142	1.00–1.02	–	23	31	124	G	125	1.00–1.02	–
9	6	111	G	148	1.00–1.02	–	24	31	125	G	126	1.00–1.02	–
10	6	112	E	135	1.00–1.02	0–800	25	60	126	E	127	1.00–1.02	0–800
11	6	113	G	136	1.00–1.02	–	26	60	127	G	142	1.00–1.02	–
12	114	135	G	135	1.00–1.02	–	27	50	128	G	148	1.00–1.02	–
13	115	135	E	136	1.00–1.02	0–800	28	72	129	E	135	1.00–1.02	0–800
14	58	116	G	125	1.00–1.02	–	29	72	130	G	136	1.00–1.02	–
15	58	117	G	126	1.00–1.02	–							